**Question 1**

1)

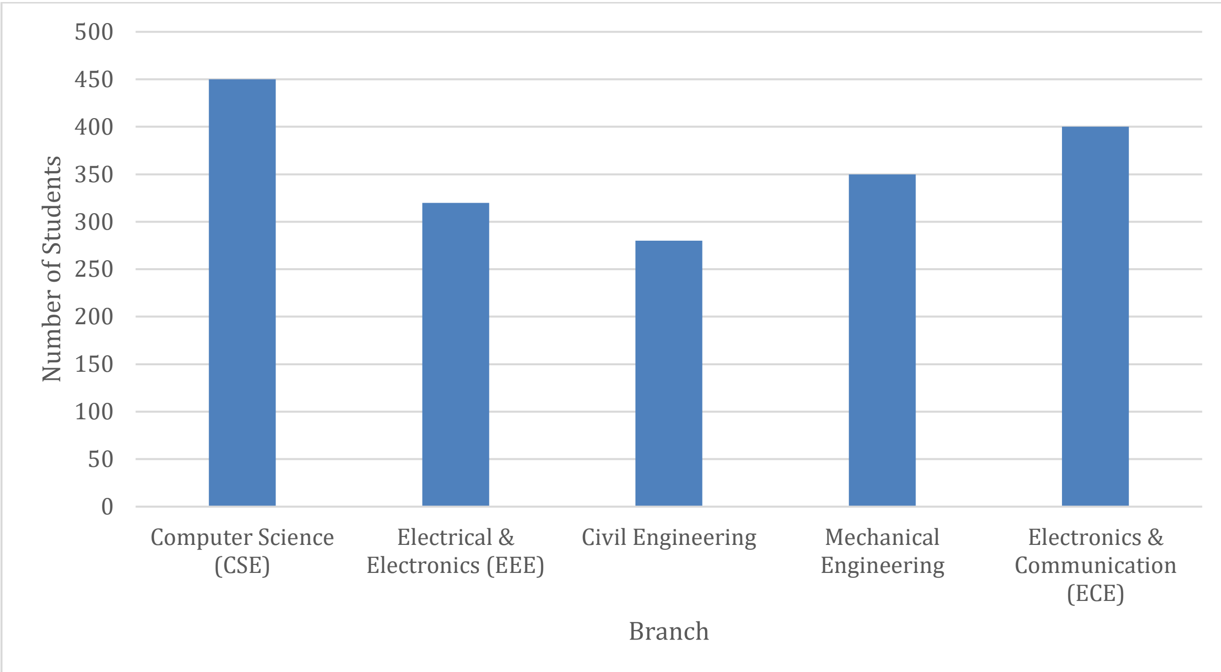



Figure 1: Bar Chart

2) To draw a pie chart, we need to calculate the percentage of the data.

Table 1: Frequency distribution table of the number of students enrolled in different engineering branches in a university

| **Branch** | **Number of Students** | **Percentage** |
|---|---|---|
| Computer Science (CSE) | 450 | 25 |
| Electrical & Electronics (EEE) | 320 | 17.78 |
| Civil Engineering | 280 | 15.56 |
| Mechanical Engineering | 350 | 19.44 |
| Electronics & Communication (ECE) | 400 | 22.22 |
| Total | n = 1800 | 100 |

Pie chart representing above data:

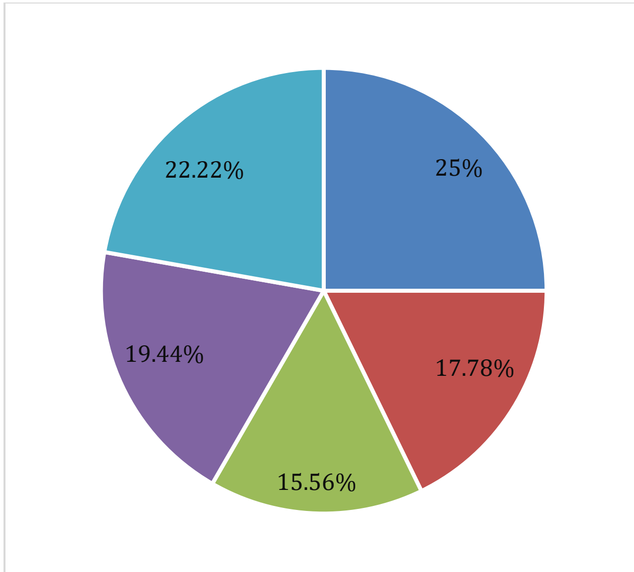



Figure 2: Pie Chart

3) The branch with the highest enrollment is Computer Science (CSE) with 450 students, and the branch with the lowest enrollment is Civil Engineering with 280 students.

## Question 2

1) The given class intervals are discrete. To draw a histogram, we need to convert them into continuous class intervals.

Table 2: Frequency distribution of the monthly electricity consumption (in kWh) of 50 households.

| **Consumption Range (kWh)** | **Number of Household** |
|---|---|
| 0 – 50.5 | 5 |
| 50.5 – 100.5 | 8 |
| 100.5 – 150.5 | 12 |
| 150.5 – 200.5 | 10 |
| 200.5 – 250.5 | 7 |
| 250.5 – 300.5 | 5 |
| 300.5 – 350.5 | 3 |

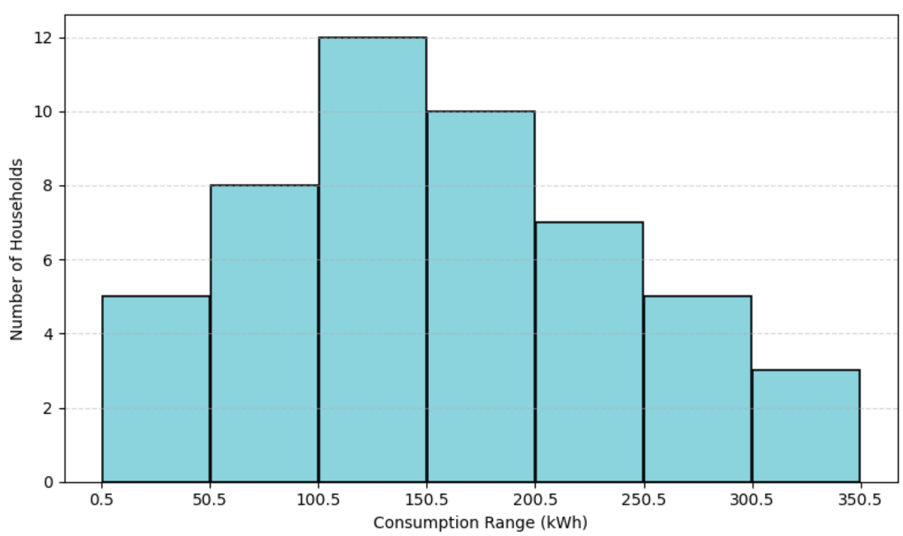
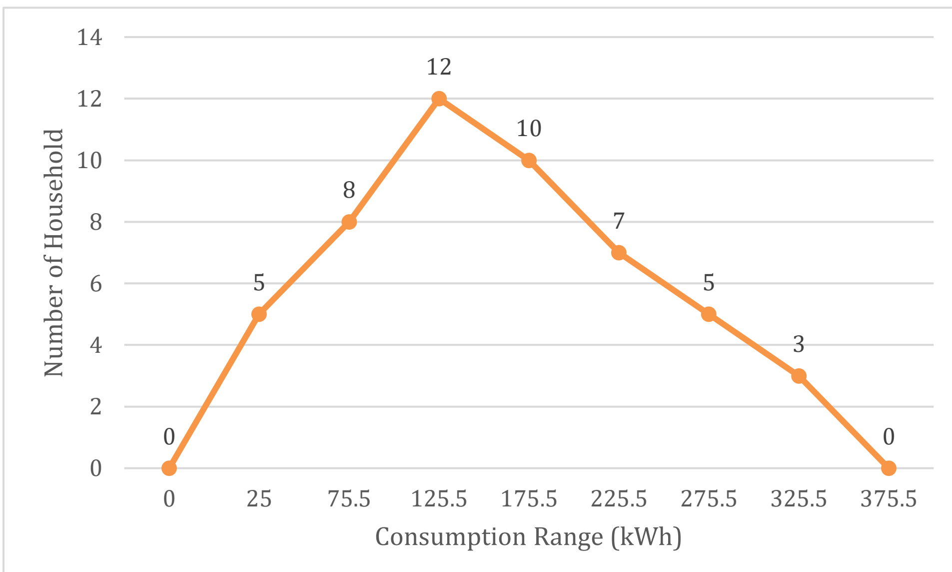



Figure 3: Histogram chart

2) To draw frequency polygon from the given data, we first determine the midpoint of each class interval.

Table 3: Frequency distribution of the monthly electricity consumption (in kWh) of 50 households.

| **Consumption Range (kWh)** | **Midpoint** | **Number of Household** |
|---|---|---|
| 0 – 50 | 25.25 | 5 |
| 51 – 100 | 75.5 | 8 |
| 101 – 150 | 125.5 | 12 |
| 151 – 200 | 175.5 | 10 |
| 201 – 250 | 225.5 | 7 |
| 251 – 300 | 275.5 | 5 |
| 301 – 350 | 325.5 | 3 |
| **Total** | | n = 50 |




**Consumption Range (kWh)**

Figure 4: Frequency Polygon

3) From the histogram and frequency polygon, we can see that most households use between 100–150 kWh of electricity. After that, the number of households slowly decreases as consumption increases.

This means the data is not balanced on both sides, it has a long tail on the right side. So, the distribution is positively skewed.

**Question 3**

1)

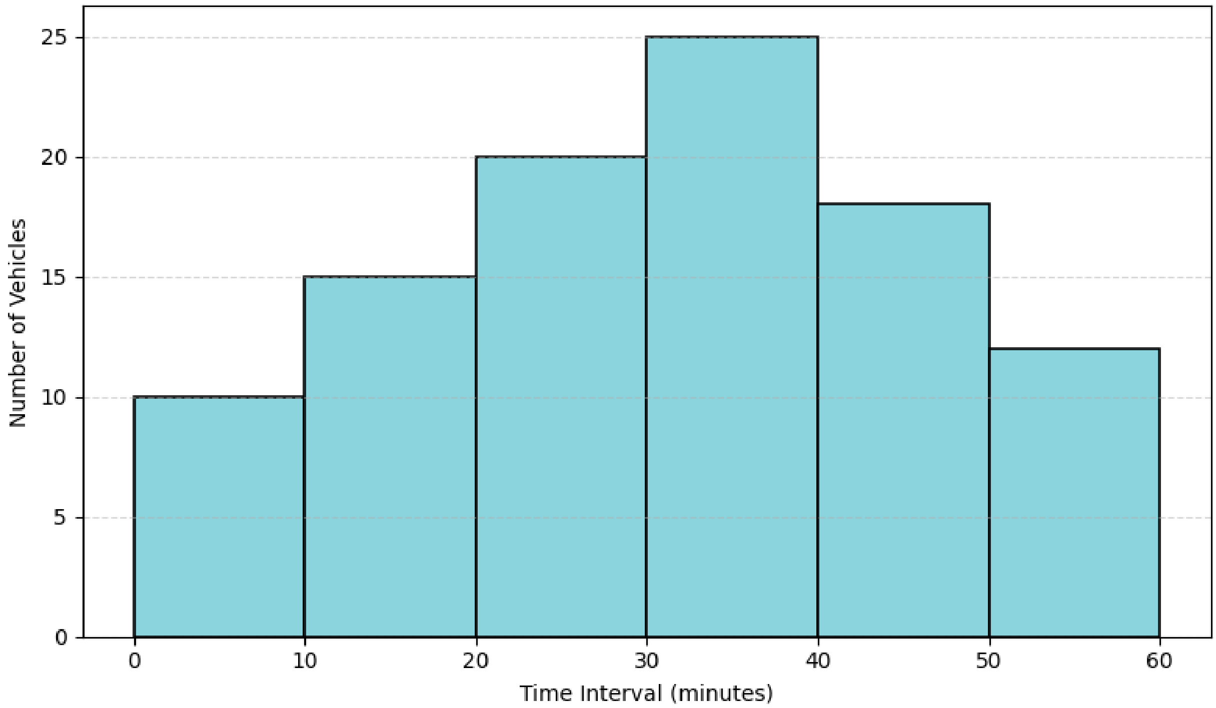



Figure 5: Histogram

2) To draw an ogive graph, we need to calculate the cumulative frequency of the data.

Table 4: Frequency distribution of the number of vehicles passing through a highway toll booth in different time

| **Time Interval (Minutes)** | **Number of Vehicles (f)** | **Cumulative Frequency (CF)** |
|---|---|---|
| 0-10 | 10 | 10 |
| 10-20 | 15 | 25 |
| 20-30 | 20 | 45 |
| 30-40 | 25 | 70 |
| 40-50 | 18 | 88 |
| 50-60 | 12 | 100 |
| Total | N = 100 | |

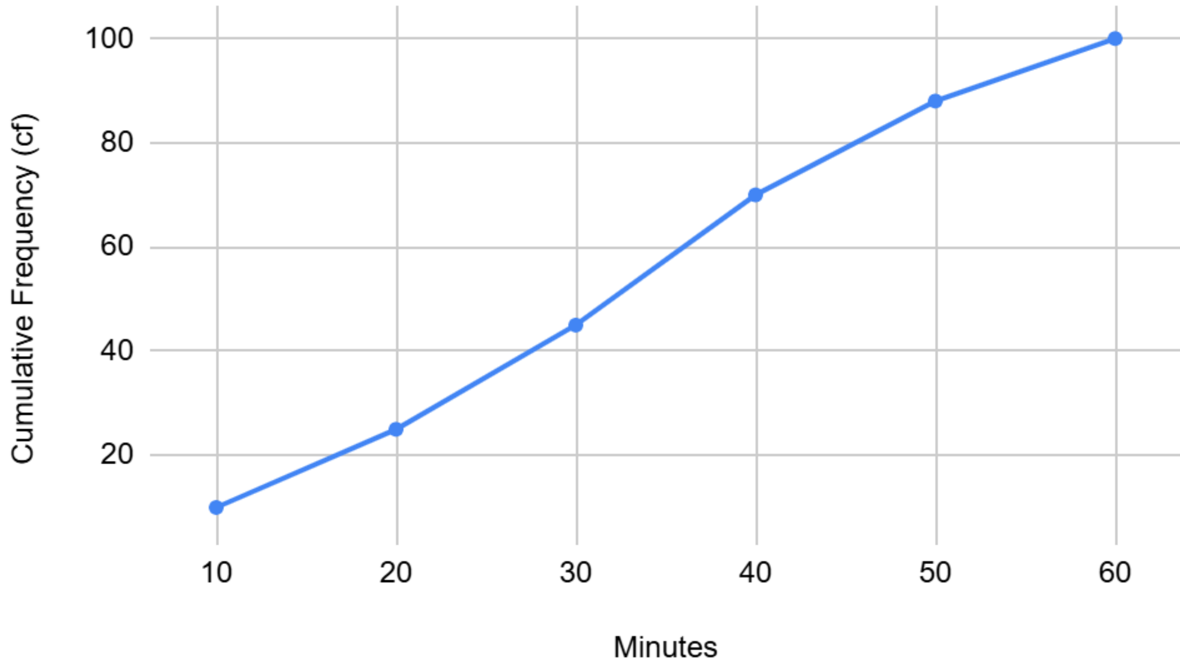



Figure 6: Ogive Curve

3)

The total number of vehicles is N=100.

Median = L + $\frac{\frac{N}{2} - p.c.f}{f} \times$ i

The median class is the value corresponding to the CF ≥ $(\frac{N}{2})^{th}$ observation, which is$(\frac{100}{2})^{th} = 50^{th}$ observation.

So, the median class is 30-40

Median = 30 + $\frac{\frac{100}{2} - 45}{25} \times 10$

= 32

The estimated median traffic count interval is 32 minutes. This means that for half of the observed time, 50 or fewer vehicles passed, and for the other half, 50 or more vehicles passed

## Question 4

1)

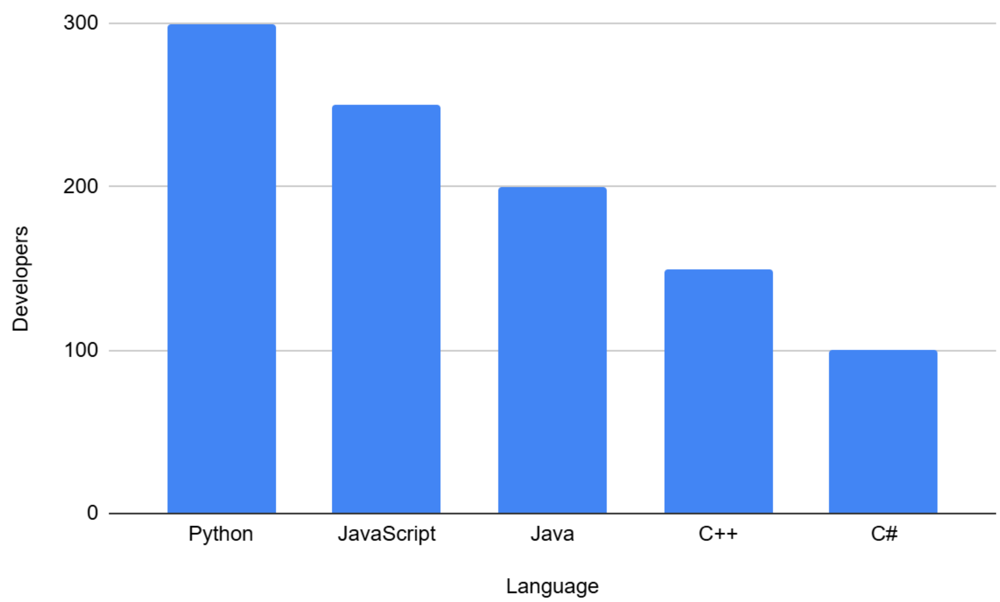



Figure 7: Bar Chart

2) To draw a pie chart, we need to calculate the percentage of the data.

Table 5: Frequency distribution of the number of software developers about their preferred programming languages

| **Language** | **Developers** | **Percentage** |
|---|---|---|
| Python | 300 | 30 |
| JavaScript | 250 | 25 |
| Java | 200 | 20 |
| C++ | 150 | 15 |
| C# | 100 | 10 |
| Total | n = 1000 | 100 |

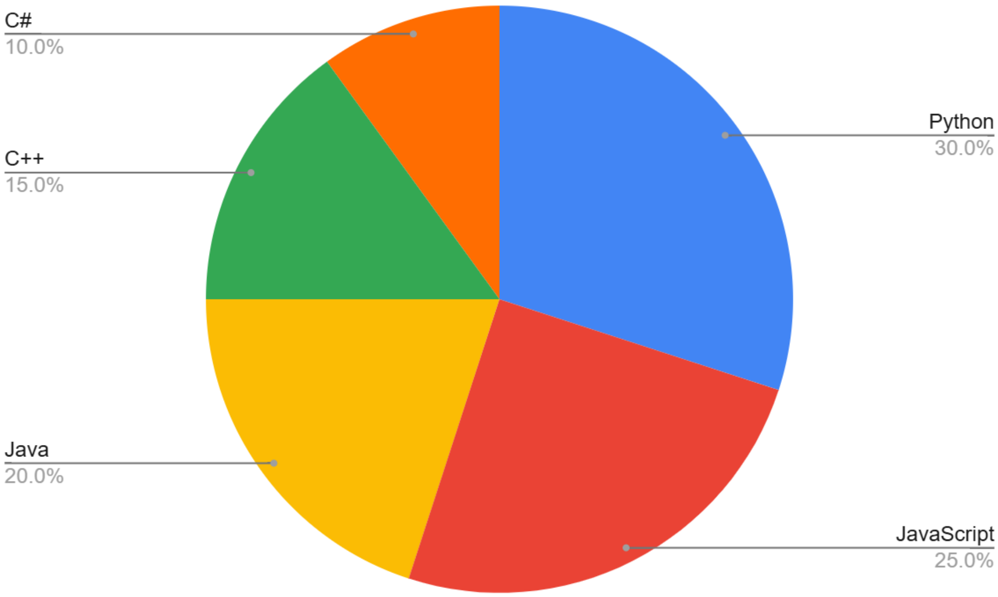



Figure 8: Pie Chart

3) Total developers = 300 + 250 + 200 + 150 + 100

= 1000

Percentage of python = $\frac{Each\ Value}{Total\ Developer} \times 100$

$= \frac{300}{1000} \times 100 = 30\%$

Percentage preferring Python: 30%